

Niagara 4 SMS & Calls Driver For SMSEagle Devices

Release 3.3

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Changelog

- Version 1.0 Initial release
- Version 2.0 Changed to APIv2, added calls (ring calls, text-to-speech calls)
- Version 3.0 Added SmsEagleAlarmRecipient – routes Niagara alarms to SMS with native %alarmData.*% token support
- Version 3.1 Added Call Duration parameter – allows to configure ringing duration of a call
- Version 3.2 Fixed bug with alarm text resolution
- Version 3.3 Add alarm-recipient TTS dispatch modes and async device dispatch

Driver Overview

The PROXIMUS Driver for SMSEagle devices is written using Tridium's BFramework, for use in Niagara 4 or later.

The driver's main purpose is to serve as a gateway to send SMS or create automatic calls (ring call, text-to-speech-call) via SMSEagle for Tridium Supervisor 4 or JACE8000, JACE9000 stations (and OEM variations i.e. Centra Line, Trend, ...).

SMSEagle is an offline hardware SMS gateway. Therefore, no external connection to 3rd party system is required. All notifications are generated on-premise and sent directly to a cellular network. This solution can be used in secure installations without Internet access.

Setup is easy and should take 10-15 min to complete.

Prerequisites for Integration

1. Correctly set up SMSEagle device (see chapter "SMSEagle Setup").
2. Correctly installed modules with the driver (see chapter "Driver Installation").

Installation & Configuration

SMSEagle Configuration

1. Create a new user in SMSEagle (menu Users > + Add Users).
2. Grant API access to the created user:
 - a. click **Access to API** beside the newly created user
 - b. Enable **APIv2**
 - c. **Generate new token** (copy it for later use)
 - d. Add access permissions in section **APIv2 Permission** for: **Send SMS**
 - e. Add access permissions in section **APIv2 Permission** for: **Make a ring call, Make a TTS Call, Make a TTS Advanced call (for voice alerting)**
 - f. **Save** settings

Driver Installation

Extract the **smsEagle_driver*.zip** archive and copy all included *.jar files to your Niagara modules directory, typically **C:\Niagara\Niagara-4.x.xx.xx\modules**.

For correct behavior, it is necessary to install *.jar files on the client platform (Workbench PC) as well as on the target platform (JACE8000/9000 or Supervisor).

Note: Close the Niagara Workbench after inserting all the modules in the folder. Next time you start the Niagara Workbench the driver will be loaded in Niagara Workbench and will be immediately available for use.

Driver Configuration

1. Connect JACE 8000/9000 or PC with Tridium Supervisor 4 to the network.
2. Add new SmsEagleNetwork to your Drivers node.
3. Under network add new SmsEagleDevice.
4. Configure the newly added device:

SmsEagleDevice (Sms Eagle Device)

Status	{down}
Enabled	☒ true
Fault Cause	
Health	Fail [17-Jun-26 9:15 AM BST] Unable to cc...
Alarm Source Info	Alarm Source Info
Poll Frequency	Normal
Http Config	Sms Eagle Http Comm Config
Fault Cause	
Use Tls	☐ false
Address	local:80
Ip Address	IP address or domain of your SMSEagle device
Port	☐ unspecified port (80 or 443) [-1 - 65536]
Connection Timeout	0 ms
Api Key	API key from SMSEagle
To	
Group	
Priority	0
Tts Voice Id	2
Tts Mode	Tts Advanced
Text	test
Result	

Fig 1: SmsEagleDevice configuration

Fill in the following properties of SmsEagleDevice

- **Enabled:** set to True
- **Use Tls:** set to True if you will use HTTPS connection (otherwise set to false)
- **Ip Address:** fill in IP address or domain name of your SMSEagle device
- **Port:** 80 (for HTTP connection) or 443 (for HTTPS connection)
- **Api Key:** API key created in SMSEagle webGUI
- **Tts Voice Id:** set to 2 (for English voice text-to-speech calls)
- **Call Duration:** allows to configure a ringing duration for a voice call

Note: if you want to use HTTPS connection between Niagara and SMSEagle you must have SSL certificate correctly configured on SMSEagle device.

Properties of SmsEagleDevice

SmsEagleDevice	
Sms Eagle Device	
Status	[down]
To	123
Group	321
Priority	0
Tts Voice Id	2
Tts Mode	Tts Advanced
Text	test
Send	
Send Group	
Ring Call	
Ring Group Call	
Tts Call	
Tts Group Call	

Every device has following properties:

- **To:** recipient phone number (or multiple numbers separated with comma)
- **Group:** (optional) group name defined in SMSEagle Phonebook. If you define the group in SMSEagle phonebook (it must be set as public), you may send SMS to the group of recipients
- **Priority:** (optional) priority of SMS – value between 0-9 (9 is the highest priority)
- **Tts Mode:** (optional) selects which endpoint TTS calls use – Tts Advanced (default; supports language selection on SMSEagle and natural sounding language) or Tts (legacy, English only, but faster processing).
- **Tts Voice Id:** (optional) numeric identifier of the TTS voice to use, as defined in the SMSEagle webGUI > menu Calls. Applies to Tts Advanced calls only. For English set: 2.
- **Text:** text of SMS message

Every device has defined the following actions:

- **Send:** send SMS to recipient defined in To parameter
- **Send Group:** send SMS to the group defined in Group parameter
- **Ring Call:** make a call to recipient defined in To parameter
- **Ring Group Call:** make a call to the group defined in Group parameter
- **Tts Call:** send SMS as a speech to recipient defined in To parameter
- **Tts Group Call:** send SMS as a speech to the group defined in Group parameter

Adding the SmsEagleAlarmRecipient

- Open the smsEagle palette in Niagara Workbench.
- Drag SmsEagleAlarmRecipient onto the station's AlarmService (or a recipient folder beneath it).
- In the Alarm Class whose alarms should be sent as SMS, add the recipient to its route so routed alarms reach the component.

Example Configuration

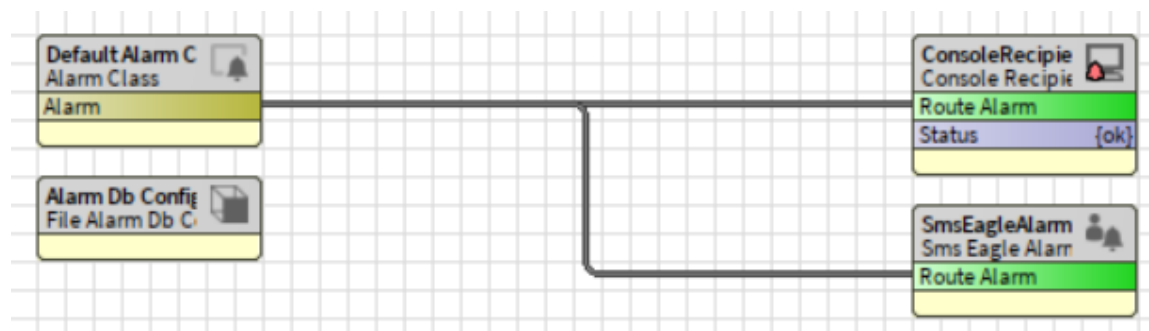


Fig 2: SmsEagleAlarmRecipient on an Alarm Class route

Plug the SmsEagleAlarmRecipient component directly into Niagara's AlarmService. This routes alarms to a configured SmsEagleDevice as an SMS, with native resolution of Niagara %alarmData.*% alarm format tokens. This produces clean alarm text automatically.

The component does not communicate with the SMSEagle hardware directly. It reuses an existing SmsEagleDevice for all connection and recipient settings (IP address, API key, To, Priority). Configure the target SmsEagleDevice as described in the previous chapters before using the recipient.

To send the source name and alarm message for every alarm of an Alarm Class:

- Add a SmsEagleAlarmRecipient under AlarmService and add it to the Alarm Class route.
- Set Sms Eagle Device Ord to your configured SmsEagleDevice.
- Set Body Format to: %alarmData.sourceName%: %alarmData.msgText%

A resulting SMS could read: "AHU-1 Supply Temp: High temperature alarm".

Properties of SmsEagleAlarmRecipient

 SmsEagleAlarmRecipient (Sms Eagle Alarm Recipient)	
 Time Range	12:00 AM - 12:00 AM
 Days Of Week	☒ Sun ☒ Mon ☒ Tue ☒ Wed ☒ Thu ☒ Fri ☒ Sat
 Transitions	☒ toOffnormal ☒ toFault ☒ toNormal ☒ toAlert
 Route Acks	☒ true
 Enabled	☒ true
 Sms Eagle Device Ord	module://smsEagle/module.palette slot:/SmsEagleDevice
 Dispatch Mode	Sms
 Body Format	<pre>%alarmData.msgText%</pre>
 Status	

Fig 3: SmsEagleAlarmRecipient property sheet

The component has the following properties:

- **Enabled:** master enable/disable. When set to false, the component ignores all routed alarms. Default: true.
- **Sms Eagle Device Ord:** an Ord pointing to the target SmsEagleDevice (already configured with IP address, API key, To, etc.). The selector lists only SmsEagleDevice instances. If the Ord does not resolve to a valid SmsEagleDevice, the alarm is ignored and a warning is written to the station log.
- **Dispatch Mode:** property with four options:
 - Sms - send an SMS to the device's To (default)
 - Tts Call - place a TTS call to the device's To
 - Tts Group Call - place a TTS call to the device's Group
 - Sms And Tts Call - both an SMS and a TTS call to the device's To
- **Body Format:** multi-line template for the SMS text, using Niagara alarm format tokens that are resolved against each incoming alarm record. Default: %alarmData.msgText%.
- **Status:** read-only indicator of the link to the device. Shows OK when the Ord resolves to a live SmsEagleDevice, or "Not linked to a SmsEagleDevice" otherwise (empty, wrong type, or a device that has been removed). It is re-evaluated when the Ord changes and on each routed alarm.

When an alarm is routed to the component, it resolves the Body Format template against the alarm record and sends the resulting text through the referenced SmsEagleDevice, using that device's To, Priority and connection settings. The recipient phone number is taken from the device's To property.

Supported Alarm Tokens

Body Format supports all %alarmData.*% tokens provided by Niagara's standard alarm token resolver. The most commonly used are:

- %alarmData.msgText% – alarm message text
- %alarmData.sourceName% – name of the alarm source
- %alarmData.priority% – alarm priority
- %alarmData.timestamp% – time the alarm occurred
- %alarmData.sourceState% – source state (e.g. offnormal, normal)
- %alarmData.ackState% – acknowledgement state

Literal text may be freely combined with tokens.